BHASWATH VENKATA V. DATLA

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EDUCATION

University of Washington — **Seattle, WA**
B.S. in Computer Science — *Expected June 2028*

GPA: 3.52 | Honors: Dean's List (Autumn 2024 - Present)

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (OOP), Discrete Mathematics, Linear Algebra, Artificial Intelligence, Database Systems (SQL/Relational Databases)

TECHNICAL EXPERIENCE

Husky Coding Project (ASL App), UW — **Seattle, WA**
Junior Programmer — *January 2025 - January 2026*

- Developed real-time ASL gesture recognition features using TensorFlow, validating model performance through a 50-person testing round that informed a 20% reduction in prediction latency.
- Resolved 15+ critical software defects and conducted weekly code reviews using AI-assisted development in Google's agentic Antigravity IDE, accelerating debugging cycles and improving application stability.

UW Sensors, Energy, and Automation Laboratory (UW SEAL) — **Seattle, WA**
Lab Associate – Gas Leak Autonomous Sensing & Detection (DOE-funded) — *March 2026 - Present*

- Built FFT-based signal-processing pipelines in Python and MATLAB to extract vibration/acoustic features from sensor data, enabling reliable leak-signature detection for a DOE-funded compressed-gas leak localization project.
- Developing a PyTorch neural network to classify leak signatures, currently labeling and structuring raw vibration telemetry to build the foundational training dataset.

TECHNICAL PROJECTS

Network Latency Observability Engine — **Personal Project**
Creator — *December 2025 - May 2026*

- Engineered a multithreaded background service in Java using java.net.Socket to continuously monitor network stability, tracking TCP connection latency and packet loss across multiple external servers.
- Managed asynchronous data ingestion utilizing thread-safe ConcurrentLinkedQueue structures, safely buffering 10,000+ daily telemetry points to local CSV logs while eliminating thread-blocking I/O operations.
- Streamlined network diagnostics by building an automated reporting module to identify latency patterns, reducing manual connectivity troubleshooting time by approximately 40%.

Red Bull AI Impactathon — **Hackathon Project (UW)**
AI & Pitch Strategist — *March 2026*

- Won 3rd Place out of a competitive field of 50+ university teams at the Red Bull AI Impactathon, collaborating within an agile 3-person team to conceptualize a high-impact software design under tight constraints.
- Architected the application's core logic and prompt engineering framework, designing context schemas and system instructions to optimize LLM reasoning and output quality.
- Delivered the final strategic presentation to a panel of industry judges, clearly communicating technical frameworks, market viability, and solution architecture.

LEADERSHIP & SERVICE

Wave Hopes and Smiles (Nonprofit)
Founding Member & Head of Development — *September 2023 - June 2025*

- Helped launch and scale a community-focused nonprofit organization, growing the team from 15+ to 20+ members and coordinating 3 major fundraising events that raised $10,000+ fully donated to Seattle Children's Hospital.
- Directed a team of 15+ volunteers while balancing rigorous academic responsibilities, demonstrating strong collaboration and communication skills, and subsequently volunteered 100+ additional hours after transitioning out of the leadership role.

ADDITIONAL SKILLS & INTERESTS

Programming Languages: Python, Java, JavaScript, HTML, SQL, PostgreSQL, SQLite, Git/GitHub.

Web & Cloud Frameworks: React, Vite, Tailwind CSS, Vercel, Firebase.

AI & Machine Learning: TensorFlow, PyTorch, scikit-learn, LLM Workflows.

AI-Assisted Development: Workflows using Agentforce, Google Antigravity, and Claude API. Skilled in LLM-assisted debugging, context optimization, and prompt engineering (Salesforce Agentblazer Champion; completing Anthropic AI Academy).

CS Fundamentals: Object-Oriented Design (OOD), Data Structures & Algorithms (DSA), Relational Database Concepts.

Independent Learning & AI Development: Member of UW RSO (Interactive Intelligence) exploring neuroscience-informed ML. Actively building and experimenting with agentic workflows and LLM coding assistants to drive rapid prototyping.